The Fertility Dividend: Family Size and the Decline in Child Poverty*

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Abstract

Child poverty under the Supplemental Poverty Measure fell 20 percentage points between 1967 and 2024. This decline is conventionally credited to income growth and changing taxes and transfers. I show that 5.4 points (27%) of the decline instead reflects a fertility dividend: smaller families spread finite resources across fewer children. Because equivalence scales and decompositions condition on family size, standard accounting records this demographic change as income growth. I quantify this effect under the assumption that the rank-order of the fertility distribution is preserved over time, and corroborate the mechanism by showing that twin-induced parity transitions raise child poverty.

Keywords: Child poverty; family size; fertility; equivalence scales; poverty measurement

JEL Classification: I32, J13, J11

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Introduction

Child poverty in the United States fell by about 20 percentage points between 1967 and 2024 under the anchored Supplemental Poverty Measure. The standard account credits growth in market income and changing taxes and transfers (Burkhauser et al. 2024; Fox et al. 2015; National Academies of Sciences, Engineering, and Medicine 2019; Wimer et al. 2016). I show that 27% of the decline—5.4 percentage points, or roughly 3.7 million fewer children in poverty in 2024—instead reflects declining fertility. As families had fewer children, resources were spread across fewer children, reducing child poverty even holding household resources fixed. The remainder reflects income growth and taxes and transfers, partially offset by rising single parenthood.

This "fertility dividend" is easy to miss in conventional poverty accounting. Analyses of poverty trends typically condition on family size (e.g., Burkhauser et al. 2024; Fox et al. 2015; Meyer and Sullivan 2012; Wimer et al. 2016), and poverty statistics convert household resources into resources per "equivalized" person using an equivalence scale whose denominator scales with family size. When family size falls, other things equal, equivalized income rises, and demographic changes are recorded as income growth rather than as a separate determinant of poverty. The resulting fertility dividend echoes the macroeconomic demographic dividend (Bloom et al. 2003; Bloom and Williamson 1998), but the mechanism operates within families: fewer children share family resources, raising resources relative to the poverty threshold. Without measuring this channel, accounts of anti-poverty progress overstate the role of income growth and understate the role of demographic change.

The key mechanism through which family size affects poverty is resource dilution, the quantity–quality channel emphasized by Becker (1960). An additional child mechanically raises a family's poverty threshold and spreads family resources across more children. Unless family resources rise by enough to offset the higher threshold, the family moves closer

to poverty. Evidence from the literature on fertility shocks provides little reason to expect such an income offset. Indeed, additional children tend to reduce mothers' labor supply (Aaronson et al. 2021; Angrist and Evans 1998).

To quantify the contribution of the fertility channel, I construct counterfactual poverty rates that replace the distribution of the number of children in one year with that of another, holding all else fixed. Within marital-status cells, I reassign to each mother in a base year the number of children at the same percentile of the target-year distribution of co-resident children. This rank-preserving transport (Novosad et al. 2022) provides a counterfactual family size and, with it, changes the poverty threshold and child exposure weights, but leaves household resources and other characteristics unchanged. I then recompute poverty under the counterfactual family-size and poverty threshold. The approach assumes that mothers retain their ordinal ranking in the distribution of fertility as that distribution shifts across years. I assess sensitivity to this assignment rule using alternative transport cells and rank orderings, and use twin-birth instruments to illustrate the key mechanism.

In the baseline specification, the results suggest that declining fertility and family size reduced child poverty by 5.4 percentage points since 1967 (about 27% of the total decline). Sensitivity analysis varying the transport specification and ranking rule yields estimates of the contribution of fertility ranging from -2.3 to -8.5 percentage points. To place this magnitude in context, I compare the fertility contribution with the effects of changes in marriage, market income, and taxes and transfers using a Shapley decomposition.

These estimates are likely a lower bound on the total effect of fertility change on child poverty because they exclude broader behavioral and compositional channels through which lower fertility may have reduced poverty, including delayed childbearing, higher parental earnings, and greater human-capital investment (e.g., Bailey 2006, 2013; Goldin and Katz 2002). To the extent that these changes raised observed family resources, they are attributed to increases in measured income in the decomposition rather than through

the family-size channel.

To provide independent evidence on the key resource-dilution mechanism driving these changes, I use twin births as an instrument for parity transitions (Angrist and Evans 1998; Rosenzweig and Wolpin 1980). Twin-induced increases in family size raise child poverty by 3–8 percentage points in recent years and 8–16 percentage points in 1960. The IV estimates also reveal the underlying causes of this relationship. Additional children raise the poverty threshold substantially, but generate little change in total family income or labor supply. This pattern supports the interpretation that the poverty effect operates through mechanical resource dilution: by changing the denominator of equivalence scales used to define the poverty thresholds.

Extending the analysis to 1900–2023 suggests that fertility was also likely a sizable contributor to poverty changes earlier in the century, a period in which average family size fell substantially.

The IV analysis suggests that the basic mechanism remains active, even if it appears no longer to be a significant driver of poverty trends in recent years. After 1980 family size among parents largely stabilized. Today's fertility decline operates increasingly through childlessness (Kearney and Levine 2025; Kearney et al. 2022), which changes the number of children rather than resources per child. However, the estimates suggest that policies that succeed in raising births among existing parents could, absent offsetting resources, mechanically raise child poverty. This finding provides a family-level analog to the observation of Weil (2026) that raising U.S. fertility back to replacement would lower living standards for several decades.

The paper makes three contributions. First, it quantifies the family-size contribution to child poverty over the past several decades. Second, it shows that standard equivalence-scale accounting attributes this demographic change to "income growth." Third, it validates the resource-dilution mechanism with an analysis of how twin births affect income and poverty.

Data

This paper draws on two microdata sources. The main analysis uses the Anchored Supplemental Poverty Measure (SPM) linked to the Current Population Survey (CPS), 1967–2024 (Flood et al. 2022; Waldfogel et al. 2017). SPM resources equal cash income plus in-kind transfers and refundable tax credits, minus taxes and "necessary" expenses. SPM Poverty Thresholds vary with family size, family composition, and geography.

For the instrumental variables analysis and the long-run historical extension, I use Decennial Census and American Community Survey microdata from IPUMS-USA (Ruggles et al. 2023). These provide the large sample sizes needed to identify twin births, and provide consistent measures of family structure over time. In these data, poverty is measured using the Official Poverty Measure (OPM), which compares pre-tax cash income to thresholds based on family size.

The sample includes all children ages 0–17 in family units containing at least one adult and one child. Each child is linked to a parent by first matching to a mother, then to a father if no mother link is available, and finally to the best-available adult in the same SPM resource unit. This hierarchy ensures that every child in the sample is linked to an adult whose characteristics (age, marital status, family income) can be used in the analysis. The SPM sample contains about 2.8 million child-year observations over 1967–2024. The Census/ACS sample contains about 3.9 million child observations spanning 1900–2023.

For the decomposition, I collapse to one observation per linked adult ("mother") and re-aggregate to children using child weights equal to the mother's survey weight multiplied by the number of children. Poverty is measured using SPM poverty and market poverty (pre-tax, pre-transfer income below the SPM threshold) to separate market-income growth from changes in transfers.

The key indicator of fertility and family size change is the number of co-resident linked children ages 0–17 (capped at seven) per mother.

For the twin-birth IV analysis, I use the 5% extract from the 1960 Census and pooled ACS samples from 2015–2024. Twin births are identified as instances in which two children of the same mother share a birth year and quarter. I exclude apparent triplets and higher-order multiples. Appendix Table E1 reports summary statistics for 1967 and 2024.

Trends in Fertility, Income, and Poverty

Declining fertility has created a growing gap between the measured economic experience of households and that of children. Figure 1 compares cumulative growth in median family income and median income per child since 1967. Median family income rose 87%, but median income per child rose 164%—nearly twice as fast. By contrast, income per equivalent person—using the scale embedded in the SPM poverty measure—grew much more slowly than income per child because of the steep economies of scale embedded in those adjustments. This wedge is one indicator of the fertility dividend. Dividing resources among fewer children raised per-child income more than indicated by household income or equivalized income. Conventional metrics may therefore understate the improvement in children's material living standards.

One illustration of the mechanism linking changes in fertility to poverty reduction is the steep and persistent gradient between poverty and family size. In 1967, about 21% of children in families with one or two children lived in poverty compared to about 41% of children in families with four or more children. This gradient appears in every year and under both SPM and OPM poverty (Appendix Figures D1 and D2).

Most of the variation in this relationship is explained by differences in family size rather than differences in income (or, specifically, that lower-income parents have more children). To see this, consider a counterfactual in which children were assigned to families at random and poverty rates re-calculated based on observed resources and (counterfactual) family size. In that scenario, the poverty–family size gradient would be about the same. For instance, in 1967, 21% of children in one- or two-child families would be poor under the

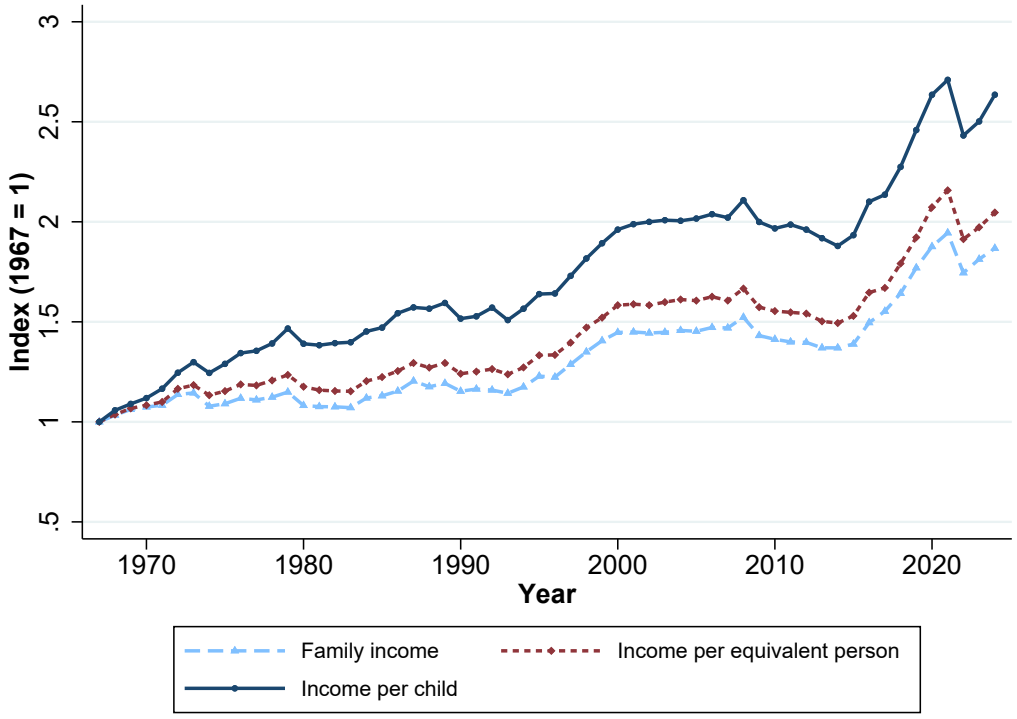


Figure 1: Income per child outpaced family income as families shrank, 1967–2024

Notes: Figure plots cumulative growth in three income measures among families with children, indexed to 1967. Family income (dashed) measures total SPM resources. Income per equivalent person (short-dashed) applies the Betson three-parameter equivalence scale used in the SPM (Betson 1996). Income per child (solid) divides family income by the number of children. The divergence between family income and income per child reflects the fertility dividend. The gap between income per child and the equivalized series measures the per-child gains that equivalence scaling conceals.

SPM in this scenario, and, among families with four or more children, 41% would be poor (Appendix Figure D2). (Controlling for other differences between families of different sizes through reweighting would not change that relationship much.)

The magnitude of the family-size–poverty gradient is large. Being in a large family is a relative risk factor for child poverty comparable to having an unmarried parent or a less-educated mother. Over the twentieth century, the share of children living in families with four or more children collapsed from over half to just 15%, shifting a large fraction of the child population from higher-poverty large families to lower-poverty small families.

The decline in fertility was also associated with other behavioral changes like later childbearing and increases in maternal education, labor force participation, and earnings, which also reduced child poverty (Bailey 2013). In the analysis below, I hold household resources fixed and thus attribute these behavioral channels to income growth rather than demographic change.

The cross-sectional association between family size and child poverty could reflect factors other than a causal effect of differences in the number of children, such as systematic differences in income and other characteristics across families with different fertility, or changes in income associated with changes in family size. The decomposition and the twin-birth IV analysis address this issue in complementary ways. The decomposition constructs counterfactual poverty rates that hold the income distribution fixed while varying family size, isolating the mechanical "resource dilution" component. The IV analysis uses twin births to generate quasi-random changes in family size, testing whether and how an additional child raises poverty.

Decomposition of Poverty Change

I construct counterfactual scenarios that ask: what would child poverty have been if families had had fewer (or more) children, but household resources stayed the same? To that end, I replace each year's family-size distribution with that of another year and recompute

poverty, holding resources fixed. Using these counterfactual poverty rates, I decompose the observed reduction in child poverty between 1967 and 2024—approximately 20 percentage points—into contributions from fertility, marriage, market income, and taxes and transfers. The counterfactual-assignment problem is how to attach the target-year family-size distribution to the reference-year population of mothers and resources. The aggregate decline in family size (and co-resident children) is observed, but it is not obvious to which mothers to assign those counterfactual co-resident children. Because the poverty effect depends on that assignment—whether children are assigned to higher- or lower-resource families—I impose a rank-preservation assumption: within marital-status cells, mothers who are relatively high-fertility in one year remain relatively high-fertility in another year. This allows fertility to be correlated with income and other family characteristics. The key assumption is that ordinal fertility ranks are stable within cells.

For example, if the 80th-percentile married mother has four children in 1967 but three in 2024, in the counterfactual, I assign the 1967 mother three children and adjust her poverty threshold for the smaller family (via the implied change in equivalence scales). I then recalculate the poverty rate for the three (not four) children exposed to that new poverty threshold estimating the national child poverty rate. This counterfactual changes only the number of co-resident children (and thus poverty thresholds and child exposure weights), but not family resources, marital status, geographic threshold adjustments, or any other household characteristic.

Because family size is discrete, within-size ranks are not identified. In the main specification, I break ties randomly, and report bounds on tie-breaking methods using income-ascending and income-descending tie-breaking orderings, following the partial-identification logic in Manski (2003).

Formally, within marital-status cell m, I compute each mother's weighted percentile p_i in the reference-year distribution of the number of children and assign the counterfactual number of children $C_i^{\text{cf}} = \min\{c : F_m^{t_1}(c) \geq p_i\}$, where $F_m^{t_1}$ is the target-year weighted

empirical CDF. I then recompute child poverty using the counterfactual threshold

$$T_i^{\text{cf}} = T_i \times \frac{\text{scale}(A_i, C_i^{\text{cf}})}{\text{scale}(A_i, C_i)}, \tag{1}$$

where $\text{scale}(A, C) = (A + 0.5C)^{0.7}$ for $A \geq 2$ and $(1.3 + 0.5C)^{0.7}$ for $A = 1$ (single-adult units with children). This ratio-based rescaling changes only the family-size component of the threshold. The average child poverty rate is calculated using counterfactual child weights equal to the mother's survey weight times C_i^{cf}.

This procedure is a rank-preserving distributional transport (Novosad et al. 2022). Let θ_i denote an unobserved, time-invariant fertility propensity for mother i and let X_i index the marital-status cell. The identifying assumption is that the conditional distribution of θ_i is stable over time:

$$\Pr(\theta_i \leq \theta \mid X_i = x,\, t_0) = \Pr(\theta_i \leq \theta \mid X_i = x,\, t_1) \quad \text{for all } \theta, x. \tag{2}$$

Under this assumption, a mother at percentile p of the family-size distribution within cell x in t_0 is also at percentile p in t_1, so that matching on percentile ranks recovers the counterfactual distribution of co-resident children. In plain English: mothers who tend to have more children than their peers in one year would also tend to have more children in another year. This assumption is weaker than assuming fertility is independent of income because it preserves any existing relationship between fertility rank and other household characteristics. High-fertility mothers may be poorer, for example, so long as their relative fertility ranks are preserved as the distribution shifts lower. I assess robustness to alternative ranking methods and tiebreaking rules below. The twin-birth instruments provide corroborating evidence that does not rely on rank stability.

To assess the magnitude of the fertility contribution, I compare it with changes in marriage, market income, and taxes and transfers using a Shapley decomposition (Shapley 1953; Shorrocks 2013; Young 1985). This approach attributes the total change in poverty

by averaging each factor's marginal contribution across alternative orderings. For each ordering, I start from the reference-year population and update one component at a time, recomputing child poverty after each step.

Fertility is updated using the rank-preserving transport described above. To estimate the contribution of the marriage component, I reweight the base year data to match the target-year child-weighted marriage share. After fertility and marriage are applied, the remaining change in poverty is allocated between market income and taxes and transfers by comparing market poverty with total poverty under the same size-adjusted threshold. Averaging each factor's marginal contribution yields components that sum to the total change in child poverty.

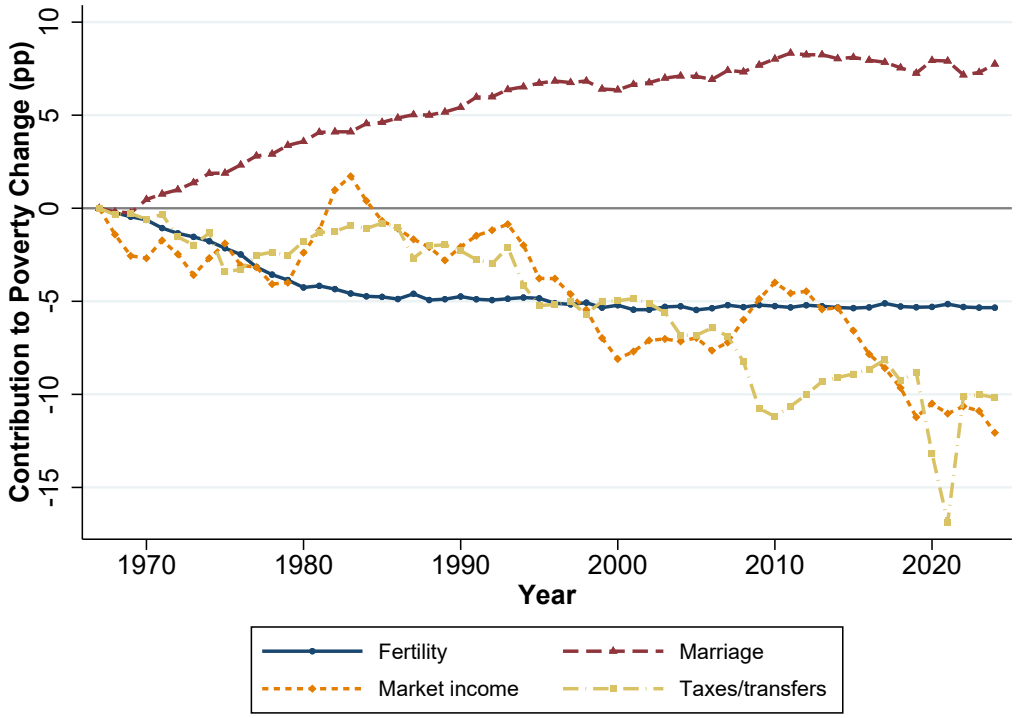


Figure 2: Fertility decline accounts for 27% of the 1967–2024 fall in child poverty

Notes: Figure plots cumulative Shapley contributions of each factor to the change in child poverty (SPM) relative to 1967. The fertility component captures changes in family size within marital-status cells; components sum to the total change by construction.

Figure 2 presents the cumulative results of the decomposition. The decline in fertility reduced child poverty by 4.3 percentage points between 1967 and 1980 alone—when family

size fell fastest—reaching −5.4 points by 2024, 27% of the total decline. The contribution of fertility was essentially complete by 2000 and flat thereafter.[1]

About 90% of the fertility contribution reflects resource dilution rather than selection by income. Having fewer children lowers the poverty threshold facing a given family and therefore makes children in the family less likely to be poor. Only 10% of the fertility contribution reflects reweighting arising from the share of children reallocated away from lower-income, higher-poverty families and toward higher-income families.

Marriage decline increased poverty by 7.7 pp, while market income growth contributed −12.0 pp and transfer expansion −10.2 pp. Within groups, the decline in fertility among married mothers accounts for roughly half of the poverty effect. Unmarried mothers contribute disproportionately relative to their population share in 1967.

Table 1 reports the fertility contribution from the full Shapley decomposition under alternative transport designs and tiebreaking rules. In every specification, poverty is decomposed into fertility, marriage, market income, and taxes and transfers as above. What varies across rows and columns is only the way the counterfactual distribution of fertility is assigned. The rows vary the conditioning cells and the treatment of fertility timing. The main specification (A) transports the number of children within marital-status cells. Method B applies an unconditional transport that ignores marital status (so there is only one cell) and assigns counterfactual fertility only based on the overall percentile distribution of children. Method C adds fertility timing by first transporting age at first birth to the target-year distribution of first birth, and then transporting the number of children within cells defined by marital status and quintile of age at first birth. A+IPW provides a timing-adjusted version of the main specification by first reweighting the reference-year sample to match the target-year age-at-first-birth distribution and then transporting the

[1]The decomposition totals reflect mother-level aggregation on the estimation sample, which differs slightly from raw child-level rates because of slight differences in within-family sample weights. The decomposition sums to −19.8 pp (baseline 29.4%) while the simple child-level difference is −20.1 pp (29.7% → 9.6%). The Shapley baseline of −5.4 pp averages the fertility contribution over orderings. The single-transport A estimate in Table 1 (−5.3 pp) isolates fertility alone and differs from the Shapley number only through the marriage-ordering interaction.

number of children within marital-status cells. These variants are intended to examine whether the fertility contribution depends on holding marriage fixed, and whether it reflects fewer children rather than later first births.

The columns vary the tiebreaking rule for how mothers are ordered within family-size units, where the rank-ordering is not otherwise identified. Because many mothers share the same observed number of children, their latent fertility ranks are not observed. In the main specification above, ties are broken at random (column 2). The income-descending and income-ascending tiebreakers assign these ranks in the two extreme orderings with respect to income. They thereby provide lower and upper bounds on the estimated fertility contribution. The copula tiebreaker uses a Gaussian copula to impose the observed within-cell rank correlation between income and family size. Appendix B describes the construction of these specifications in detail.

Table 1: Sensitivity of the estimated fertility contribution to child poverty

	Tiebreaker			
Transport method	Inc-desc.	Random	Inc-asc.	Copula
A: Within marriage	−2.9	−5.3	−8.5	−5.6
B: Marginal	−2.3	−5.0	−8.1	−5.2
C: Sequential AFB → fert	−2.5	−4.9	−7.6	−5.9
A+IPW: Marriage + AFB	−3.8	−6.1	−8.2	−6.6

Notes: Table reports the fertility-transport contribution to child poverty (percentage points, 1967–2024) under four conditioning specifications and four within-tie ordering rules. Income-descending assigns counterfactual children to higher-income mothers first (lower bound); income-ascending assigns to lower-income mothers first (upper bound). The single-transport estimate for the main specification (A, random tiebreaker) is −5.3 pp; the −5.4 pp figure cited in the text is the Shapley fertility contribution, which adds the marriage-ordering interaction. Source: CPS ASEC, 1967–2024.

Across all 16 method×tiebreaker combinations, the estimated contribution ranges from −2.3 to −8.5 percentage points, with the central estimates (random and copula tiebreakers) clustering between −4.9 and −6.6 pp. The copula tiebreaker produces estimates close to the random tiebreaker because the income–fertility gradient within cells is relatively flat.

The final row of Table 1 (A+IPW) is a timing-adjusted variant of the main transport.

Before transporting family size within marital-status cells, I reweight the reference-year sample using inverse probability weights so that the age-at-first-birth distribution matches that of the target year, and then recompute counterfactual poverty. Relative to A, this adjustment increases the estimated fertility contribution by about 1 percentage point. This calculation uses the reference year's incomes, age– earnings profile, and returns to education and work, but it is not a causal estimate of how delayed childbearing affected earnings or human capital. Rather, it is intended to show that shifting the age-at-first-birth distribution toward its later target-year pattern might be expected to result in additional poverty reduction even before assigning any behavioral earnings gains to delayed fertility. Expanding the transport cells to race × marital status yields similar estimates (Appendix B).

The estimated fertility contribution is also robust to alternative poverty measures. Appendix Table B2 recomputes the transport effect under square-root, per-capita, deep-poverty, and fixed per-child thresholds, and under the two equivalence scales economists have estimated directly from expenditure data—Lazear and Michael (1980) and Nelson (1988), which embed the largest economies of scale in the literature (Fernández-Villaverde and Krueger 2007). The Betson and square-root scales yield nearly identical estimates (−5.0 and −4.9 pp). The consumption-based scales bracket this benchmark (−6.0 pp for Lazear–Michael, −3.2 pp for Nelson). And the per-capita measure produces a larger effect (−8.2 pp). The central estimate is not an artifact of the SPM's assumed equivalence scale.

The estimated fertility contribution illustrated in Table 1 should be interpreted primarily as a mechanical resource-dilution effect. It captures how smaller families raise resources per child. This effect is economically meaningful, as it is roughly half the size of the entire contribution of taxes and transfers (−10.2 pp). Moreover, the decomposition underestimates the effect of fertility on poverty because it attributes to "income growth" the broader behavioral channels through which fertility control changed behavior and composition, such as delayed childbearing, higher educational attainment, increased labor force

participation, and higher earnings (e.g., Bailey 2006, 2010, 2013; Goldin and Katz 2002).

Instrumental Variables Estimates of Family Size and Poverty

The decomposition above holds family resources fixed and finds that declining family size contributed to falling poverty primarily because fewer children share those fixed resources. To assess this resource-dilution interpretation, I use twin births as instruments for parity transitions (Angrist and Evans 1998; Rosenzweig and Wolpin 1980). This evidence addresses the concern that the poverty–family-size gradient partly reflects selection into larger families. Conditional on a pregnancy at a given parity, a twin birth generates a plausibly exogenous increase in family size, and thus provides evidence of how that change affects family income, labor supply, and the family's poverty threshold.

I define parity transitions based on observed co-resident children ages 0–17. Using the 1960 Census and pooled 2015–2024 ACS data, I estimate the effect of crossing each parity threshold—going from one child to two, from two to three, and so on—by instrumenting for an indicator for having at least $n+1$ co-resident children with a twin birth at pregnancy n (Angrist et al. 2010; Black et al. 2005). In the baseline sample, I restrict to mothers aged 30 and older with at least n pregnancies, to approximate completed fertility. (Appendix Table C4 shows the estimates are nearly identical when the sample is widened to all mothers aged 18 and older.) The first stage is strong, with F-statistics exceeding 3,000.

Table 2 presents the results. Panel A shows that twin-induced parity transitions increase the probability that a child lives in poverty by 3–8 percentage points in the modern period and 8–16 percentage points in the 1960 Census, when baseline poverty rates were higher. Why? Panel B shows that total family income is largely unaffected by twin-induced family-size increases, and Panel C shows that earnings per child fall sharply. At the same time, however, an additional child raises the family's official poverty threshold by 15–36 log points across margins (Panel D).

The resource-dilution mechanism is apparent in the contrast across Panels B and D: the

Table 2: Twin-IV mechanism test: effect of crossing a parity threshold

	Parity Transition					
	$1 \to 2$	$2 \to 3$	$3 \to 4$	$4 \to 5$	$5 \to 6$	$6 \to 7$
Panel A: Child Poverty (OPM)						
ACS	0.031	0.037	0.043	0.038	0.079	0.065
	(0.004)	(0.004)	(0.009)	(0.016)	(0.037)	(0.063)
	[< .001]	[< .001]	[< .001]	[.017]	[.033]	[.306]
Census	0.099	0.084	0.095	0.126	0.143	0.158
	(0.011)	(0.010)	(0.013)	(0.018)	(0.026)	(0.033)
	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]
Panel B: Log Family Income						
ACS	0.115	0.014	−0.011	0.039	−0.154	0.043
	(0.010)	(0.010)	(0.020)	(0.032)	(0.092)	(0.128)
	[< .001]	[.173]	[.595]	[.225]	[.093]	[.738]
Census	0.017	−0.031	−0.026	−0.008	−0.060	−0.103
	(0.020)	(0.018)	(0.022)	(0.031)	(0.050)	(0.066)
	[.408]	[.082]	[.234]	[.784]	[.233]	[.119]
Panel C: Log Earnings per Child						
ACS	−0.731	−0.459	−0.340	−0.272	−0.332	−0.234
	(0.011)	(0.011)	(0.021)	(0.040)	(0.092)	(0.140)
	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]	[.096]
Census	−1.111	−0.629	−0.490	−0.409	−0.271	−0.328
	(0.030)	(0.023)	(0.030)	(0.041)	(0.058)	(0.087)
	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]
Panel D: Log Poverty Threshold (OPM)						
ACS	0.232	0.188	0.147	0.149	0.198	0.241
	(0.002)	(0.002)	(0.003)	(0.006)	(0.015)	(0.020)
	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]
Census	0.363	0.245	0.204	0.208	0.218	0.232
	(0.006)	(0.005)	(0.005)	(0.007)	(0.010)	(0.009)
	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]	[< .001]
Obs. (ACS)	2,644,308	1,511,086	507,798	146,809	42,508	14,887
Obs. (Census)	921,213	622,001	338,051	165,357	77,334	37,721

Notes: 2SLS estimates. Each column instruments an indicator for having at least n+1 children with a twin birth at pregnancy n. Panel A shows that twin-induced parity transitions raise child poverty; Panel B shows that total family income is largely unaffected; Panel C shows that earnings per child fall proportionally; Panel D reports the (mechanical) effect on the log official poverty threshold, isolating the change in the needs standard induced by an additional child. The positive income effect at the $1 \to 2$ margin in the ACS likely reflects ART-related selection; this anomaly does not appear in the 1960 Census (which predates ART) or at higher parities. Controls include maternal age, education, marital status, survey year, and age-at-first-birth fixed effects. Sample restricted to mothers aged 30+. ACS = pooled American Community Survey, 2015–2024; Census = 1960 Decennial Census. OLS comparisons are in Appendix C. Heteroskedasticity-robust standard errors in parentheses; continuous p-values in brackets.

needs standard rises sharply while family resources do not—so measured poverty rises. Nor is there an offsetting increase in parental labor supply at other margins (Appendix C.7). More generally, this pattern—the poverty threshold rises, income is little changed, resources per child fall, poverty rises—is a useful description of why declining fertility has tended to reduce poverty over the period examined in the decomposition.

One anomaly is at the $1 \to 2$ margin in modern data, where family income rises, perhaps reflecting selection among mothers using assisted reproductive technology. This anomaly is absent in the 1960 Census and at all higher parities. (Full regression results including OLS comparisons are reported in Appendix C.)

The IV estimates are also broadly consistent with the magnitude implied by the decomposition. Mean family size fell from 3.3 to 2.4 children between 1967 and 2024 (a decline of 0.9), implying a substantial shift in births away from higher parities. For comparison, scaling the modern parity-specific IV estimates (roughly 3–8 percentage points across margins) by the 0.9-child decline implies a poverty reduction of roughly 3–7 percentage points, which brackets the decomposition's estimate of 5.4 percentage points. The larger 1960 IV estimates reflect the fact that the density of families clustered near the poverty threshold was greater in 1960. When more families sit just above the line, an additional child tips a larger share below it. This is a consequence of the density of the (equivalized) income distribution around the threshold, not of the higher baseline poverty rate per se. Together, the IV results indicate that the poverty–family size gradient reflects resource dilution rather than selection, and support the decomposition's estimates.

Long-Run Perspective: 1900–2023

The CPS–SPM series begins in 1967—after much of the twentieth-century fertility transition—and therefore misses decades in which U.S. family size fell more sharply than in the post-1967 period. To put the main results in historical context, I extend the analysis to Decennial Census and ACS microdata spanning 1900–2023 (IPUMS USA), using the Official

Poverty Measure (OPM) data available in those sources. This extension quantifies the mechanical implications of fertility change during 1900–1967, when shifts in family size were larger than in the CPS era. An added benefit is that it tests whether the same resource-dilution channel appears under the OPM poverty measure both before and after the SPM measure becomes available.

Figure 3 compares decade-by-decade decompositions from the CPS–SPM analysis (panel a) and the Census/ACS–OPM analysis (panel b). Panel a reports the above Shapley contributions from fertility, marriage, market income, and taxes/transfers under the SPM. Panel b reports fertility and marriage components under the OPM, with a residual that reflects growth in cash income (because taxes and non-cash transfers are not used in the OPM).

In panel a, fertility's poverty-reducing effect is concentrated early in the CPS window. During 1967–1980, when average family size fell rapidly, fertility alone is estimated to have reduced child poverty by 4.3 percentage points, exceeding even the contribution of market income growth. After 1980, fertility's marginal contribution is small, consistent with the deceleration in family-size decline. Market income growth drives much of the poverty decline in the 1990s (-5.8 pp) and 2010s (-8.0 pp), while transfer expansion is the primary factor in the 2000s (-6.1 pp). (Appendix Table A1 reports the full numerical results; Figure A1 provides a complementary decomposition of income per child.)

Panel b applies the same fertility transport within marital-status cells and the same reweighting by marriage used in Panel a using the Census OPM data. The residual is attributed to cash income growth. Because OPM poverty is not available before 1950, the 1900–1950 bars are the implied effects of backward-transporting the 1950 baseline income and poverty thresholds to earlier distributions of family sizes.

In the years that they overlap, the OPM-based estimates mirror those using the SPM. Fertility's contribution is concentrated in the late 1960s and 1970s and is comparatively small after 1980. Across two poverty concepts and two data sets, the decomposition assigns the

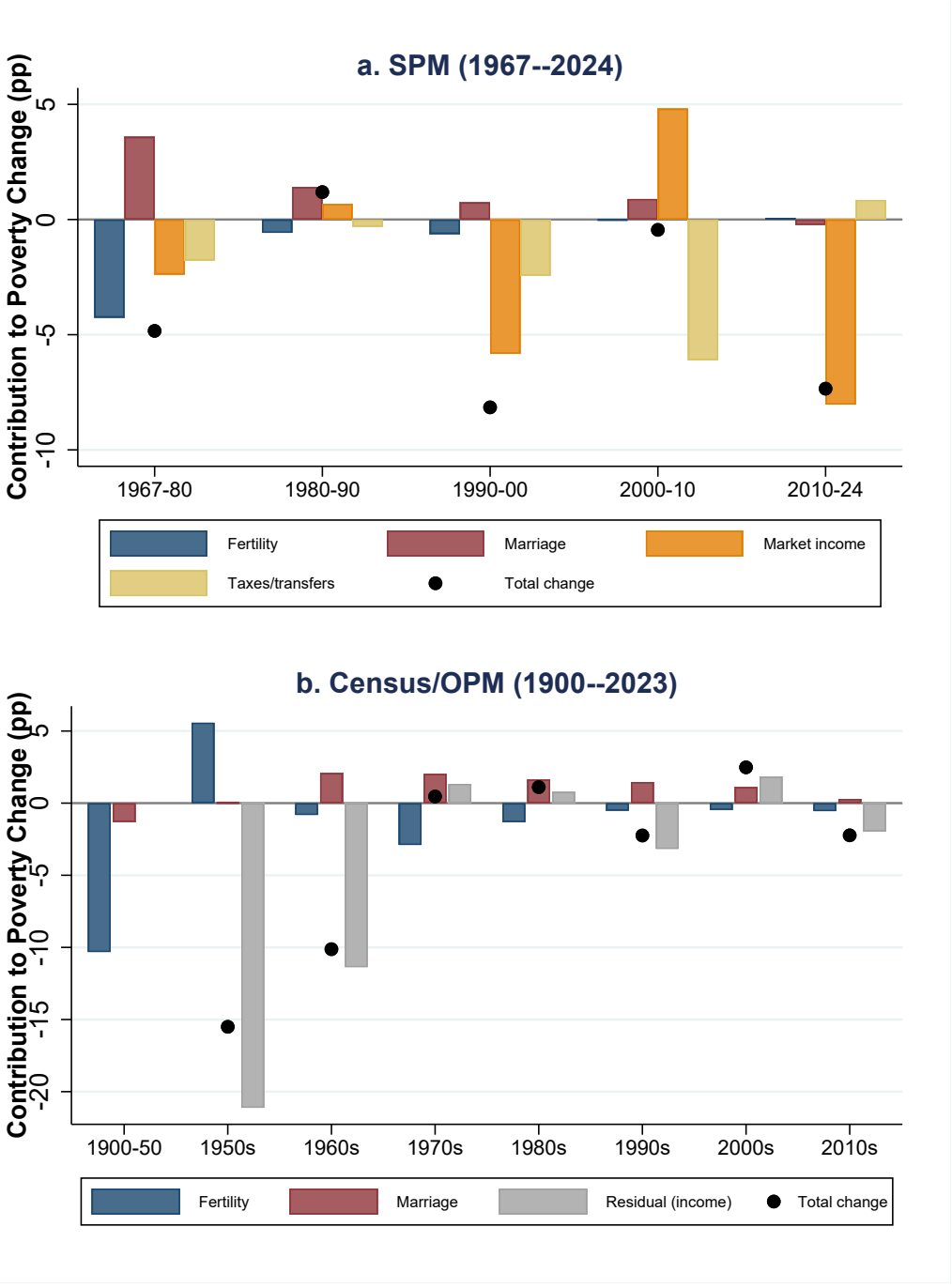


Figure 3: Fertility's contribution to falling child poverty was concentrated before 1980

Notes: Panel a decomposes changes in SPM child poverty (1967–2024) into Shapley contributions from fertility, marriage, market income, and taxes/transfers. Panel b decomposes changes in OPM child poverty (1950–2023) into fertility (rank-preserving transport), marriage (IPW reweighting), and a residual capturing income and policy changes. OPM poverty is available from 1950 onward. The 1900–1950 fertility and marriage bars are computed by applying a backward transport to the 1950 baseline—the total change and residual are not observed for this period. Black dots show total observed poverty change. Sources: CPS ASEC (panel a), IPUMS USA Census/ACS (panel b).

fertility dividend to the same historical period when family-size decline was fastest, and the cumulative contribution is similar.

The long-run series also suggests that fertility likely played an even larger role before the CPS window. Indeed, the largest implied fertility effect comes in the period before 1950, when mean family size fell from about 4.2 to roughly 3 children—a decline at least as large as the subsequent fall to 2.4. The 1950s baby boom temporarily reversed the family-size decline and rising family size appears to have mechanically pushed child poverty upward before the secular decline resumed.

Conclusion

Declining family size was a substantial and overlooked contributor to the postwar fall in child poverty. Between 1967 and 2024, smaller families raised resources per child and reduced child poverty by roughly 5.4 percentage points, accounting for 27% of the decline. Estimates using twin births suggest that resource dilution is the mechanism. Extending the analysis to Census data spanning 1900–2023 shows that the same channel was likely important earlier in the century.

The result improves child-poverty accounting in two ways. First, it shows that conventional decompositions can attribute demographic change to income growth. Because poverty measures equivalize resources and standard decompositions condition on family size, falling family size raises measured resources relative to needs without appearing as a distinct demographic contribution. This paper measures the impact of the decline in the number of children sharing family resources on poverty and attributes that component to demographic change rather than to income growth.

Second, the channel remains active even though its contribution to recent trends is small. The twin estimates show that an additional child raises a family's child-poverty risk by 3–8 percentage points in recent data. Changing fertility appears to have had little effect on child poverty in recent decades largely because family size has been relatively stable

after 1980. Today's fertility decline operates increasingly through childlessness and fewer first births (Kearney and Levine 2025; Kearney et al. 2022), a margin that changes the number of children rather than the resources available to children with siblings. However, policies that raise births among existing parents could raise child poverty by roughly the magnitudes estimated here.

Finally, the findings add a child-focused perspective to debates about low fertility and population aging (e.g., Spears and Geruso 2025; Weil 2026). Lower fertility may strain the financing of pay-as-you-go retirement programs, but increasing fertility will raise the number of dependents long before it raises the number of workers, a macroeconomic transition cost emphasized by Weil (2026). The results here show the same demographic transitions may also affect resources available for children and child poverty.

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